

Definition

- 1 Suppose $A = (a_{ij})$ is an $n \times n$ matrix. Then the submatrix A_{ij} is obtained by removing the i^{th} row and j^{th} column from A . The (i, j) -cofactor is

$$C_{ij} := (-1)^{i+j} \det(A_{ij}).$$

- 2 We define the determinant inductively: if A is an $n \times n$ matrix and $1 \leq i \leq n$ then

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}.$$

Equivalently, for any $1 \leq j \leq n$,

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}.$$

$$A = \begin{pmatrix} \textcircled{1} & \textcircled{-1} & \textcircled{2} \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$

$$A_{11} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$$

$$A_{12} = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$$

$$A_{13} = \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix}$$

$$\begin{aligned} C_{11} &= (-1)^{1+1} \det(A_{11}) \\ &= 1 \end{aligned}$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$C_{12} = (-1)^{1+2} \det(A_{12})$$

$$= (-1) (-5) = 5$$

$$C_{13} = (-1)^{1+3} \det(A_{13})$$

$$= -2$$

$$\det(A) = 1 \cdot C_{11} + (-1) C_{12} + 2 C_{13}$$

$$= 1 - 5 - 2 \cdot 2 = \underline{\underline{-8}}.$$

Properties of the determinant

Theorem

The determinant is well-defined i.e. any of the $2n$ ways of computing the determinant from the definition are equal.

Compute $\det(A)$, A as before, down the second column.

$$\begin{pmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$

$$C_{12} = (-1)^{1+2} \det \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$$

$$= 5$$

$$C_{22} = (-1)^{2+2} \det \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$C_{32} = (-1)^{3+2} \det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = -3$$

$$\det(A) = (-1) \cdot 3 + 1 \cdot (-5) + \underline{0 \cdot 2} = \underline{-8}$$

Properties of the determinant

Theorem

Suppose that B is obtained from A by a single row operation.

- 1 *If the row operation is interchanging two rows the $\det(B) = -\det(A)$.*
- 2 *If the row operation is multiplying a row by k then $\det(B) = k \det(A)$.*
- 3 *If the row operation is adding a multiple of one row to another then $\det(B) = \det(A)$.*

Theorem

For any A , $\det(A) = \det(A^T)$.

$$\begin{array}{c} \uparrow \\ n \\ \downarrow \end{array} \begin{pmatrix} * & & & \\ 0 & * & & \\ 0 & & * & \\ \vdots & & & \ddots \\ 0 & \dots & 0 & * \end{pmatrix} \leftarrow \begin{array}{l} \text{row-echelon} \\ \text{matrix.} \end{array}$$

$\leftarrow n \rightarrow$

Ex: $A = \begin{pmatrix} 1 & -2 & 3 & 4 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. What is $\det(A)$?

$$\begin{aligned} \det(A) &= 1 \cdot \det \begin{pmatrix} 1 & -3 & 2 \\ 0 & 1 & 1 \end{pmatrix} = 1 \cdot 2 \det \begin{pmatrix} -3 & 2 \\ 0 & 1 \end{pmatrix} \\ &= \underline{\underline{1 \cdot 2 \cdot (-3) \cdot 1}}
 \end{aligned}$$

$$\det \begin{pmatrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix} = \det \begin{pmatrix} 4 & 0 & 6 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$

(add R_2 to R_1).

$$= \det \begin{pmatrix} 4 & 6 \\ 2 & 1 \end{pmatrix}$$

$$= -8$$

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix} = 1 \cdot 2 \cdot 3 = 6$$

$$\begin{aligned} \det \begin{pmatrix} 0 & 2 & 1 \\ 1 & 2 & 3 \\ 0 & 0 & 3 \end{pmatrix} &= 3 \cdot \det \begin{pmatrix} 0 & 2 \\ 1 & 2 \end{pmatrix} \\ &= 3 \cdot (-2) = -6. \end{aligned}$$

Properties of the determinant

Theorem

$$\det(AB) = \det(A) \det(B).$$

Theorem

$\det(A) \neq 0$ iff A is invertible.

Properties of the determinant

Theorem

If $v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_n$ are $(n - 1)$ vectors in $\mathbb{R}^n$ then the map $T : \mathbb{R}^n \rightarrow \mathbb{R}$ given by

$$T(x) = \det(v_1, \dots, v_{j-1}, x, v_{j+1}, \dots, v_n)$$

is a linear transformation.